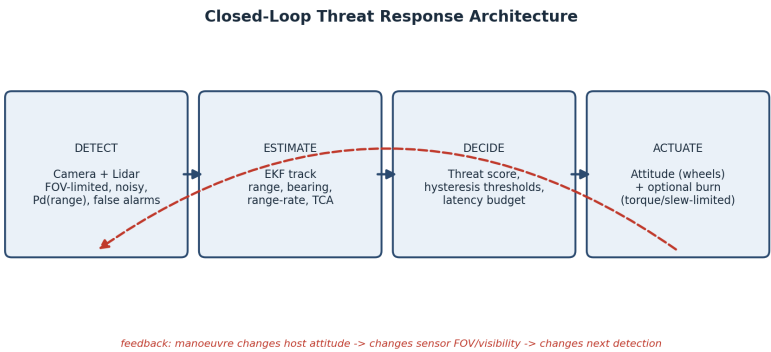


Closed-Loop Threat Response Simulation — Design Note

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Scope note. This is a **2D prototype**, built, run and debugged in Python; it is not the full 3D design. Differences are listed in Section 2. All numbers below are from the prototype.

1 Architecture

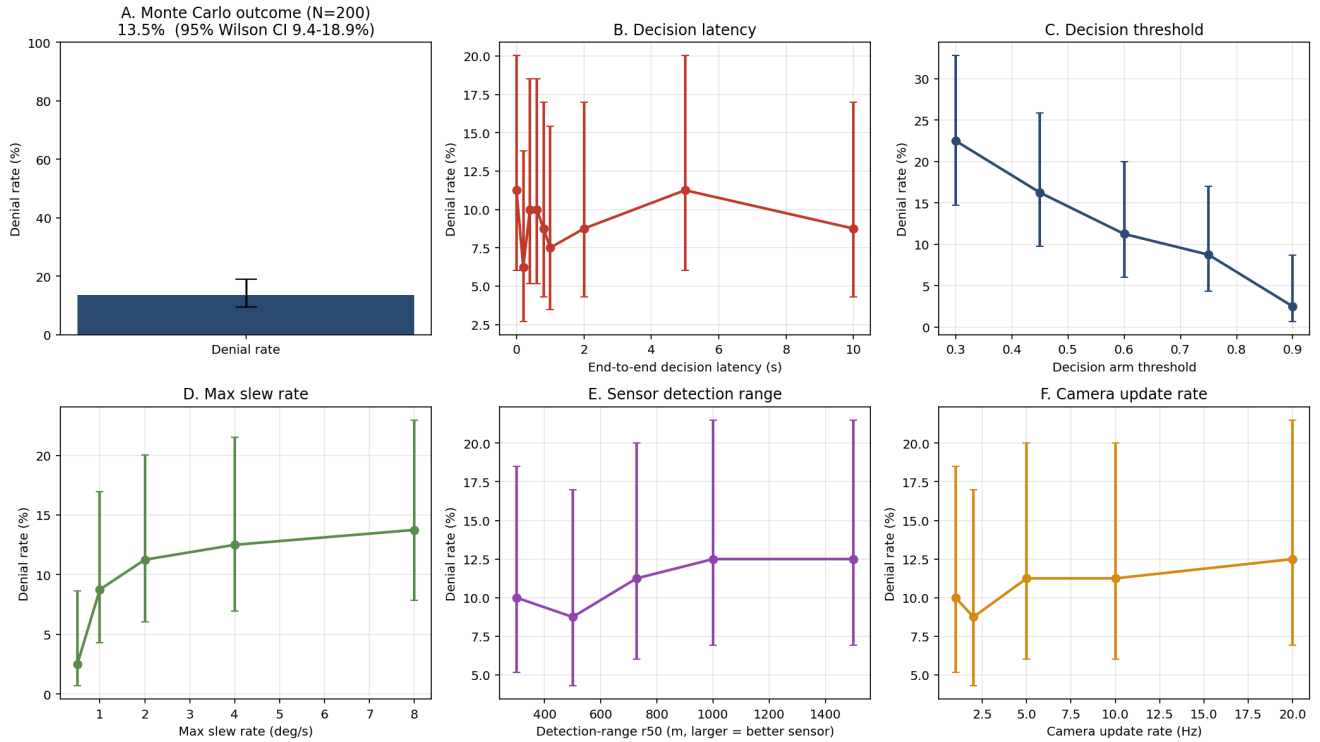


The manoeuvre that defends the host also turns its camera, since both are modelled on one body axis: once the chaser is more than $\sim 30^\circ$ off boresight (half the 60° field of view, FOV), it leaves the sensor’s view. Keeping the chaser in view (needed to keep sensing it) therefore points the vulnerable docking port at it — the central coupling this exercise is built to surface.

2 Assumptions and simplifications

Part	Designed	Implemented / defended
Relative motion	3D Clohessy-Wiltshire (CW)	Planar CW is implemented and validated against its closed-form solution (error $< 10^{-4}$ m over 500 s), but the Monte Carlo chaser instead flies a commanded straight-line path to an aim point on the port axis with a braking profile; only its re-aiming manoeuvres are charged to its 1.0 m/s budget. Separation (300–800 m) is small vs. orbital radius; 30 min is $\sim 1/3$ of the 92.6-min period, long enough that CW’s linearisation is not separately checked against a nonlinear model here.
Estimator	Extended Kalman filter (EKF)	Alpha-beta filter (gains $\alpha = 0.05$, $\beta = \alpha^2/(2 - \alpha)$, tuned so range-rate error, 0.03–0.12 m/s, stays well below the 0.3–1 m/s closing speeds being measured); no covariance or statistical gating. A track needs 3-of-5 consistent detections and 5 s of age before it can arm.
Sensor	Camera 5 Hz + lidar 2 Hz, FOV-gated	Detection probability is a logistic curve solved exactly from the brief’s anchors ($P_d(200) = 0.95$, $P_d(800) = 0.40 \Rightarrow r_{50} = 727.4$ m, $k = 0.00558$ m $^{-1}$), an empirical surrogate, not a first-principles optical model. Lidar is sampled at 5 Hz (not 2 Hz) and only returns a range when the camera also detects. False alarms: 2% per camera frame.
Actuation	Quaternion, 3-axis reaction-wheel array	Single body angle; ideal torque/rate-limited source (0.01 N·m, $2^\circ/\text{s}$), saturated every step. No wheel-momentum model (18 N·m·s accumulates over 30 min at max torque; saturation vs. a specific wheel was not checked). Host does not translate.
Decision	Predicted miss-distance metric	Heuristic score (closing speed + time-to-closest-approach + track-age confidence), two thresholds (arm/clear) for hysteresis, explicit latency between arming and the command taking effect.

3 Results (N=200; 4–6 plots below)



Denial rate: 27/200 = 13.5% (95% Wilson CI 9.4–18.9%). Track initiation occurs at a median *true* range of only 3 m (IQR 2–9 m; confirmed in 197/200 runs) — in most runs the chaser is not seen until the final few metres, because it must start inside the fixed 60° FOV (~1 in 6 scenarios) to be seen early. Of the 38 runs where the host did arm early, 25 were denied (25 of the 27 total denials); the closed loop is demonstrably active in all 38 (own-slew FOV loss ≥ 1 in every one). Other metrics: estimation error after a 5 s-old track (range 0.22 m, range-rate 0.077 m/s RMS medians); false-alarm-triggered manoeuvres 0.000/run; chaser re-aim Δv mean 0.165 m/s; cumulative torque impulse mean 1.65, max 13.8 N·m·s.

Which parameter dominates. Paired comparisons (same 100 scenarios, two settings, exact McNemar test) give a far more reliable answer than the unpaired error bars above:

Parameter (extremes)	Denied A/B	Flipped (A only/B only)	Exact p
Decision threshold 0.3 vs 0.9	24 / 2	22 / 0	< 0.0001
Max slew rate 0.5 vs 8°/s	3 / 18	0 / 15	0.0001
Latency 0 vs 10 s	11 / 12	2 / 3	1.0
Detection range 300 vs 1500 m	12 / 15	1 / 4	0.38
Camera rate 1 vs 20 Hz	14 / 14	6 / 6	1.0

Decision threshold and max slew rate dominate; both give the host more time/authority to turn before the chaser’s budget or the clock runs out. Latency, detection range and camera rate show no detectable effect at this sample size (100 pairs resolve $\gtrsim 10$ -point effects). Upstream of all five sits the fixed FOV, which decides whether the host ever gets an early track at all. **To confirm on real hardware:** bench-characterise actual wheel torque/momentum limits and end-to-end onboard compute latency (the parameter I would expect to matter once sensing is fixed), since both were idealised here.

4 Limitations and planned improvements

The following are the areas where this prototype would most benefit from further work, in roughly the order I would prioritise them.

- **Decouple camera boresight from the docking-port axis.** Currently both are modelled as a single body angle, which means keeping the chaser in view and defending the port compete for the same variable. This is the single change I would make first, and the one most likely to change the results.
- **Revisit the latency-injection behaviour.** The brief’s self-check 4 expects added decision latency to

reduce the denial rate; in this prototype it does not, including when I tested at 10 s and with detection deliberately delayed (150 m, 80 m detection ranges gave paired $p = 0.77\text{--}1.0$). In this prototype the arming decision happens hundreds of metres out or not at all, governed by field-of-view geometry and the arming threshold rather than by seconds of delay. I would want to trace this further before relying on the latency parameter in any design conclusion.

- **Move the chaser onto real relative-motion dynamics.** The chaser currently flies a commanded straight-line path rather than being propagated with the (already implemented and validated) Clohessy-Wiltshire equations, and only its re-aiming manoeuvres are charged against its 1.0 m/s budget. 24 of the 27 denials in the Monte Carlo set are this budget running out, so the result is sensitive to that choice; tying the chaser to the validated dynamics and a more principled budget model would make the denial rate more defensible.
- **Replace the heuristic threat score with a predicted-miss-distance metric.** The current score (closing speed, time-to-closest-approach, track-age confidence) is simple and interpretable, but the sensitivity sweep shows it is itself an influential design choice. A metric based on the predicted miss distance to the docking corridor, together with the estimate's uncertainty, would be more directly tied to the actual question being asked.
- **Test against an adaptive adversary.** The reported denial rate reflects this specific host design against one fixed, non-adversarial chaser policy. Testing against a chaser that adapts to the host's behaviour would give a more meaningful measure of robustness.
- **Add statistical rigour where it is currently thin:** covariance-based gating in the estimator (rather than a fixed bearing window), a wheel-momentum model for the actuator, and larger paired samples with a test for interactions between parameters rather than one-at-a-time sweeps.